Formalization of Weingarten Calculus

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August 5, 2026

Chapter 1

Schur-Weyl Duality

1.1 Definitions and Commutants

Let $V = \mathbb{C}^d$ denote the d-dimensional complex Euclidean space (modeled as `Fin d` $\to \mathbb{C}$). We study the k-fold tensor power of V and the actions of the symmetric group S_k and the endomorphism algebra $\mathrm{End}(V)$ on it.

Definition 1.1.1 (Tensor Power Space)**.** The k-fold tensor power of V is defined as:

$$V^{\otimes k} = \bigotimes_{i=1}^{k} V \tag{1.1}$$

Definition 1.1.2 (Permutation Action)**.** Given a permutation $\sigma \in S_k$, the permutation operator W_σ on $V^{\otimes k}$ acts by permuting the tensor factors:

$$W_\sigma(v_1 \otimes \cdots \otimes v_k) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(k)} \tag{1.2}$$

Definition 1.1.3 (Diagonal Action)**.** Given an operator $g \in \mathrm{End}(V)$, the diagonal action $g^{\otimes k}$ on $V^{\otimes k}$ acts as g on each tensor factor:

$$g^{\otimes k}(v_1 \otimes \cdots \otimes v_k) = g(v_1) \otimes \cdots \otimes g(v_k) \tag{1.3}$$

Definition 1.1.4 (Permutation Image)**.** The permutation image $\mathcal{P}_{d,k} \subseteq \mathrm{End}(V^{\otimes k})$ is the set of all permutation operators:

$$\mathcal{P}_{d,k} = \{W_\sigma \mid \sigma \in S_k\} \tag{1.4}$$

Definition 1.1.5 (Diagonal Image)**.** The diagonal image $\mathcal{D}_{d,k} \subseteq \mathrm{End}(V^{\otimes k})$ is the set of all diagonal action operators:

$$\mathcal{D}_{d,k} = \{g^{\otimes k} \mid g \in \mathrm{End}(V)\} \tag{1.5}$$

Lemma 1.1.6 (Permutation Action on Simple Tensors)**.** *For any simple tensor $\bigotimes_i v_i$, we have:*

$$W_\sigma(\bigotimes_i v_i) = \bigotimes_i v_{\sigma^{-1}(i)} \tag{1.6}$$

Lemma 1.1.7 (Diagonal Action on Simple Tensors)**.** *For any simple tensor $\bigotimes_i v_i$, we have:*

$$g^{\otimes k}(\bigotimes_i v_i) = \bigotimes_i g(v_i) \tag{1.7}$$

Lemma 1.1.8 (Action Commutativity on Simple Tensors). *For any permutation $\sigma \in S_k$, operator $g \in End(V)$, and vector family v:*

$$W_\sigma(g^{\otimes k}(v_1 \otimes \cdots \otimes v_k)) = g^{\otimes k}(W_\sigma(v_1 \otimes \cdots \otimes v_k)) \tag{1.8}$$

Theorem 1.1.9 (Permutation-Diagonal Commutativity). *For any $\sigma \in S_k$ and $g \in End(V)$, the permutation action and diagonal action commute as endomorphisms:*

$$W_\sigma \circ g^{\otimes k} = g^{\otimes k} \circ W_\sigma \tag{1.9}$$

Theorem 1.1.10 (Permutation Image Centralizes Diagonal Image). *The permutation image is contained in the centralizer of the diagonal image:*

$$\mathcal{P}_{d,k} \subseteq Comm(\mathcal{D}_{d,k}) \tag{1.10}$$

Theorem 1.1.11 (Diagonal Image Centralizes Permutation Image). *The diagonal image is contained in the centralizer of the permutation image:*

$$\mathcal{D}_{d,k} \subseteq Comm(\mathcal{P}_{d,k}) \tag{1.11}$$

1.2 First Fundamental Theorem of Invariant Theory

To prove the hard direction of Schur-Weyl duality, we first prove that the centralizer of the permutation image is contained in the span of the diagonal image. This is a form of the First Fundamental Theorem (FFT) of invariant theory.

Definition 1.2.1 (Tensor Basis (Primed)). The standard basis of $V^{\otimes k}$ constructed as the tensor product of the standard basis of V:

$$(e_I)_{I \in [d]^k} \tag{1.12}$$

Definition 1.2.2 (Endomorphism Matrix Representation (Primed)). The linear equivalence between the endomorphism algebra $\mathrm{End}(V^{\otimes k})$ and the algebra of $d^k \times d^k$ complex matrices:

$$\mathrm{End}(V^{\otimes k}) \simeq \mathrm{Mat}_{d^k \times d^k}(\mathbb{C}) \tag{1.13}$$

Lemma 1.2.3 (Matrix Representation of Diagonal Action (Primed)). *For any operator $g \in End(V)$, the matrix elements of its diagonal action $g^{\otimes k}$ in the tensor basis are given by:*

$$(g^{\otimes k})_{I,J} = \prod_{m=1}^{k} g_{I(m),J(m)} \tag{1.14}$$

Lemma 1.2.4 (Permutation Action on Tensor Basis (Primed)). *The permutation action of W_σ on a basis element e_I permutes the indices:*

$$W_\sigma(e_I) = e_{I \circ \sigma^{-1}} \tag{1.15}$$

Definition 1.2.5 (Tensor Basis). The standard basis of $V^{\otimes k}$ constructed as the tensor product of the standard basis of V:

$$(e_I)_{I \in [d]^k} \tag{1.16}$$

Definition 1.2.6 (Endomorphism Matrix Representation). The linear equivalence between the endomorphism algebra $\mathrm{End}(V^{\otimes k})$ and the algebra of $d^k \times d^k$ complex matrices:

$$\mathrm{End}(V^{\otimes k}) \simeq \mathrm{Mat}_{d^k \times d^k}(\mathbb{C}) \tag{1.17}$$

Lemma 1.2.7 (Permutation Action on Tensor Basis)**.** *The permutation action of W_σ on a basis element e_I permutes the indices:*

$$W_\sigma(e_I) = e_{I\circ\sigma^{-1}} \tag{1.18}$$

Lemma 1.2.8 (Matrix of Permutation Action)**.** *The matrix entries of the permutation action W_σ are:*

$$(W_\sigma)_{I,J} = \delta_{I,J\circ\sigma^{-1}} \tag{1.19}$$

Lemma 1.2.9 (Matrix Representation of Diagonal Action)**.** *For any operator $g \in End(V)$, the matrix elements of its diagonal action $g^{\otimes k}$ in the tensor basis are given by:*

$$(g^{\otimes k})_{I,J} = \prod_{m=1}^{k} g_{I(m),J(m)} \tag{1.20}$$

Lemma 1.2.10 (DiagAction of Diagonal Operator)**.** *If $g \in End(V)$ is a diagonal operator with entries f, then $g^{\otimes k}$ is a diagonal operator with entries $\prod_{m=1}^{k} f(I(m))$.*

Lemma 1.2.11 (Monomial Coefficient Independence)**.** *If a linear combination of distinct monomials in d^2 variables over $\mathbb{C}$ evaluates to zero at all points, then all coefficients are zero.*

Definition 1.2.12 (Pair Count Coincidence)**.** For two index functions $I, J : [k] \to [d]$, the coincidence count pairCount(I, J) is a function counting how many indices m have $(I(m), J(m)) = (a, b)$ for each $(a, b) \in [d] \times [d]$.

Lemma 1.2.13 (Product Monomial Representation)**.** *The product of matrix elements can be written as a monomial evaluation of the pair count:*

$$\prod_{m=1}^{k} g_{I(m),J(m)} = \prod_{(a,b)} g_{a,b}^{pairCount(I,J)(a,b)} \tag{1.21}$$

Lemma 1.2.14 (Matrix Orbit Invariance)**.** *If an operator $X \in End(V^{\otimes k})$ commutes with all permutation operators W_σ, then its matrix entries in the computational basis satisfy $X_{I\circ\sigma,J\circ\sigma} = X_{I,J}$ for all $\sigma \in S_k$.*

Lemma 1.2.15 (Pair Count Permutation Invariance)**.** *Permuting the indices by σ preserves the pair count:*

$$pairCount(I \circ \sigma, J \circ \sigma) = pairCount(I, J) \tag{1.22}$$

Lemma 1.2.16 (Sum Grouped by Pair Count)**.** *A double sum over indices can be grouped by the coincidence type (pair count).*

Lemma 1.2.17 (Linear Functional Vanishing)**.** *If a matrix is constant on coincidence types, and the coincidence-type sums of coefficients vanish, then the total inner product sum vanishes.*

Lemma 1.2.18 (Functional Separation)**.** *If x is a vector in a finite-dimensional space V not belonging to a submodule S, there exists a linear functional $\varphi \in V^*$ vanishing on S but such that $\varphi(x) \neq 0$.*

Theorem 1.2.19 (First Fundamental Theorem)**.** *The centralizer of the permutation action is contained in the span of the diagonal image:*

$$Comm(\mathcal{P}_{d,k}) \subseteq Span(\mathcal{D}_{d,k}) \tag{1.23}$$

1.3 Double Commutant Theorem and Schur-Weyl Duality

Using representation theory, we can establish the Double Commutant Theorem for the permutation algebra.

Definition 1.3.1 (Permutation Monoid Homomorphism)**.** The permutation action as a monoid homomorphism $\rho_{\text{mon}} : S_k \to \text{End}(V^{\otimes k})$ mapping $\sigma \mapsto W_\sigma$.

Definition 1.3.2 (Permutation Representation)**.** The permutation representation is the representation $\rho : S_k \to \text{End}(V^{\otimes k})$ obtained from the permutation monoid homomorphism.

Definition 1.3.3 (Permutation Module)**.** The space $V^{\otimes k}$ viewed as a $\mathbb{C}[S_k]$-module under the permutation representation.

Definition 1.3.4 (Permutation Algebra Homomorphism)**.** The algebra homomorphism $\tilde{\rho} : \mathbb{C}[S_k] \to \text{End}(V^{\otimes k})$ induced by the representation.

Lemma 1.3.5 (Evaluation of Permutation Algebra Homomorphism)**.** *The permutation algebra homomorphism evaluated on a permutation $\sigma \in S_k$ is the permutation operator W_σ:*

$$\tilde{\rho}(\sigma) = W_\sigma \tag{1.24}$$

Definition 1.3.6 (Centralizer to EndModule Mapping)**.** The mapping that identifies any $\mathbb{C}$-linear endomorphism f in the centralizer of the span of permutation operators as a $\mathbb{C}[S_k]$-linear endomorphism of the permutation module.

Definition 1.3.7 (Double Centralizer to EndEndModule Mapping)**.** The mapping that identifies any $\mathbb{C}$-linear endomorphism T in the double centralizer of the span of permutation operators as an endomorphism over the $\mathbb{C}[S_k]$-linear endomorphism ring of the permutation module.

Lemma 1.3.8 (EndModule is in Centralizer)**.** *Any $\mathbb{C}[S_k]$-linear endomorphism of the permutation module restricts to a $\mathbb{C}$-linear endomorphism that lies in the centralizer of the span of permutation operators.*

Theorem 1.3.9 (Range of Algebra Homomorphism)**.** *The image of the group algebra homomorphism is the span of the permutation operators:*

$$Im(\tilde{\rho}) = Span(\mathcal{P}_{d,k}) \tag{1.25}$$

Lemma 1.3.10 (Membership in Span of Permutation Operators)**.** *Any operator $f \in End(V^{\otimes k})$ that lies in the range of the permutation algebra homomorphism belongs to the span of the permutation operators:*

$$f \in Im(\tilde{\rho}) \implies f \in Span(\mathcal{P}_{d,k}) \tag{1.26}$$

Lemma 1.3.11 (Semisimplicity of Permutation Module)**.** *The permutation module $V^{\otimes k}$ is a semisimple $\mathbb{C}[S_k]$-module. This is a consequence of Maschke's theorem.*

Lemma 1.3.12 (Finiteness of Permutation Module over Endomorphism Ring)**.** *The permutation module $V^{\otimes k}$ is finitely generated as a module over its $\mathbb{C}[S_k]$-endomorphism ring.*

Theorem 1.3.13 (Jacobson Density for Permutation Representation)**.** *The natural map from the group algebra $\mathbb{C}[S_k]$ to the double endomorphism ring of the permutation module is surjective.*

Theorem 1.3.14 (Double Commutant containment for Permutation Algebra)**.** *The double centralizer of the span of permutation operators is contained in the span itself:*

$$Comm(Comm(Span(\mathcal{P}_{d,k}))) \subseteq Span(\mathcal{P}_{d,k}) \tag{1.27}$$

Theorem 1.3.15 (Double Commutant for Permutation Algebra)**.** *The double centralizer of the span of permutation operators is the span itself:*

$$Comm(Comm(Span(\mathcal{P}_{d,k}))) = Span(\mathcal{P}_{d,k}) \tag{1.28}$$

We now assemble the main result.

Theorem 1.3.16 (Hard Direction of Schur-Weyl Duality for Small Dimensions)**.** *The centralizer of the diagonal image is contained in the span of the permutation image.*

Theorem 1.3.17 (Hard Direction of Schur-Weyl Duality)**.** *The centralizer of the diagonal image is contained in the span of the permutation image:*

$$Comm(\mathcal{D}_{d,k}) \subseteq Span(\mathcal{P}_{d,k}) \tag{1.29}$$

Theorem 1.3.18 (Easy Direction of Schur-Weyl Duality)**.** *The span of the permutation image is contained in the centralizer of the diagonal image:*

$$Span(\mathcal{P}_{d,k}) \subseteq Comm(\mathcal{D}_{d,k}) \tag{1.30}$$

Theorem 1.3.19 (Schur-Weyl Duality)**.** *The centralizer of the diagonal image is exactly equal to the span of the permutation image:*

$$Comm(\mathcal{D}_{d,k}) = Span(\mathcal{P}_{d,k}) \tag{1.31}$$

Chapter 2

Weingarten Calculus

2.1 Solvability of the Weingarten System

We study the moment computations of operators under the Haar measure, which reduces to solving a system of linear equations. We first analyze this system in an abstract Hilbert-space setting.

Theorem 2.1.1 (Abstract Gram-System Solvability)**.** *Let H be a finite-dimensional complex Hilbert space, $v_1, \dots, v_n \in H$, and $w \in H$. The linear system:*

$$\langle v_j, w\rangle = \sum_{i=1}^{n} c_i \langle v_j, v_i\rangle \quad \textit{for all } j \tag{2.1}$$

always has at least one solution $c \in \mathbb{C}^n$, which is given by the coefficients of the orthogonal projection of w onto $Span(v_1, \dots, v_n)$.

To apply this to operators on $V^{\otimes k}$, we define permutation operators and their adjoints.

Definition 2.1.2 (Permutation Operators)**.** The permutation operator $V_d(\sigma) \in \mathrm{End}(V^{\otimes k})$ is the linear map corresponding to $\sigma \in S_k$.

Definition 2.1.3 (Adjoint Permutation Operators)**.** The dual permutation operator $V_d^{\dagger}(\sigma)$ is defined as the permutation operator of σ^{-1}.

Theorem 2.1.4 (Hermitian Adjoint Agreement)**.** *In the computational basis of $V^{\otimes k}$, the matrix representation of $V_d^{\dagger}(\sigma)$ is the conjugate transpose of the matrix representation of $V_d(\sigma)$.*

Definition 2.1.5 (Matrix Vectorization)**.** The vectorization $\mathrm{endVec}(O)$ of an operator $O \in \mathrm{End}(V^{\otimes k})$ is the vector of its matrix entries in the computational basis, viewed as an element of the Euclidean space.

Theorem 2.1.6 (The Trace Bridge)**.** *The Euclidean inner product of the vectorizations of $V_d(\sigma)$ and O equals the trace of their product:*

$$\langle endVec(V_d(\sigma)), endVec(O)\rangle = Tr(V_d^{\dagger}(\sigma)O) \tag{2.2}$$

Theorem 2.1.7 (Solvability of Weingarten Linear System). *For any operator $O \in End(V^{\otimes k})$, the Weingarten system of $k!$ equations:*

$$Tr(V_d^\dagger(\sigma)O) = \sum_{\pi \in S_k} c_\pi Tr(V_d^\dagger(\sigma)V_d(\pi)) \quad \textit{for all } \sigma \in S_k \tag{2.3}$$

always has at least one solution $c : S_k \to \mathbb{C}$.

Lemma 2.1.8 (Adjoint Trace Linearity). *The trace against $V_d^\dagger(\sigma)$ is linear in combinations of permutation operators.*

Lemma 2.1.9 (Operator Span Inclusion). *A linear combination of permutation operators lies in their span.*

Theorem 2.1.10 (Existence of Moment Representation Operator). *There exists a linear combination $M = \sum_\pi c_\pi V_d(\pi)$ lying in $Span(\mathcal{P}_{d,k})$ whose traces match O:*

$$Tr(V_d^\dagger(\sigma)M) = Tr(V_d^\dagger(\sigma)O) \quad \textit{for all } \sigma \in S_k \tag{2.4}$$

Theorem 2.1.11 (Moment Operator Decomposition from Trace and Span Properties). *If any operator M lies in $Span(\mathcal{P}_{d,k})$ and has $Tr(V_d^\dagger(\sigma)M) = Tr(V_d^\dagger(\sigma)O)$ for all σ, then its coefficients solve the Weingarten system.*

2.2 Uniqueness and the Gram Matrix for $k \leq d$

When the number of tensor factors k is at most the dimension d, the permutation operators are linearly independent, which implies that the solution to the Weingarten system is unique.

Definition 2.2.1 (Abstract Gram Matrix). The Gram matrix of a family v is $G_{j,i} = \langle v_j, v_i \rangle$.

Definition 2.2.2 (Abstract Gram Target Vector). The Gram target vector is $u_j = \langle v_j, w \rangle$.

Theorem 2.2.3 (Abstract Gram Matrix Application). *The entries of the Gram matrix satisfy $gramMatrix(v)_{j,i} = \langle v_j, v_i \rangle$.*

Theorem 2.2.4 (Abstract Gram Target Application). *The entries of the target vector satisfy $gramVec(v, w)_j = \langle v_j, w \rangle$.*

Theorem 2.2.5 (Matrix Formulation of Gram System). *A vector c solves the Gram system if and only if $Gc = u$.*

Theorem 2.2.6 (Invertibility of Independent Gram Matrix). *If the vectors $v_1, \ldots, v_n$ are linearly independent, then their Gram matrix is invertible.*

Theorem 2.2.7 (Gram Matrix Determinant is a Unit). *If the family of vectors is linearly independent, the determinant of its Gram matrix is a unit in $\mathbb{C}$.*

Theorem 2.2.8 (Unique Gram Solution). *If the vectors are linearly independent, the Gram system has a unique solution given by $c = G^{-1}u$.*

Definition 2.2.9 (Abstract Gram Solution Set). The set of all solution vectors c to the Gram system.

Theorem 2.2.10 (Gram Solution Set is Singleton). *For a linearly independent family of vectors, the solution set of the Gram system is the singleton containing $G^{-1}u$.*

Theorem 2.2.11 (Uniqueness of Gram System Solution)**.** *For a linearly independent family of vectors, there exists a unique solution to the Gram system.*

We apply this to permutation operators.

Definition 2.2.12 (Matrix Entries Equivalence)**.** The linear equivalence between matrices and the function space of their entries.

Definition 2.2.13 (Endomorphism Vectorization Equivalence)**.** The linear equivalence between endomorphisms and the Euclidean space of their matrix entries.

Definition 2.2.14 (Linear Endomorphism Vectorization)**.** The linear map representing the vectorization of endomorphisms.

Theorem 2.2.15 (Linear Vectorization Application)**.** *The linear endomorphism vectorization matches the standard vectorization:* $endVec_\ell(O) = endVec(O)$*.*

Theorem 2.2.16 (Linear Vectorization is Injective)**.** *The linear endomorphism vectorization is injective.*

Theorem 2.2.17 (Linear Vectorization Kernel)**.** *The kernel of the linear endomorphism vectorization is the trivial submodule.*

Theorem 2.2.18 (Linear Independence of Vectorized Permutations)**.** *For* $k \leq d$*, the vectorized permutation operators* $endVec(V_d(\sigma))$ *are linearly independent.*

Theorem 2.2.19 (Linear Independence of Permutation Operators)**.** *For* $k \leq d$*, the permutation operators* $V_d(\sigma)$ *are linearly independent in* $End(V^{\otimes k})$*.*

Definition 2.2.20 (Weingarten Gram Matrix)**.** The Weingarten Gram matrix is defined by $G_{\sigma,\pi} = \mathrm{Tr}(V_d^\dagger(\sigma)V_d(\pi))$.

Definition 2.2.21 (Weingarten Target Vector)**.** The Weingarten target vector is $u_\sigma = \mathrm{Tr}(V_d^\dagger(\sigma)O)$.

Theorem 2.2.22 (Weingarten Gram as Vectorized Gram)**.** *The Weingarten Gram matrix is the Gram matrix of vectorized permutation operators.*

Theorem 2.2.23 (Weingarten Target as Vectorized Target)**.** *The Weingarten target vector is the Gram vector of vectorized permutation operators.*

Theorem 2.2.24 (Cycle Count Formula for Gram Entries)**.** *The entry* $Tr(V_d^\dagger(\sigma)V_d(\pi))$ *equals the number of index functions invariant under* $\sigma^{-1}\pi$*, which evaluates to* $d^{\#cycles(\sigma^{-1}\pi)}$*.*

Definition 2.2.25 (Natural Weingarten Gram Matrix)**.** The natural-number valued Weingarten Gram matrix whose entries are the coincidence counts.

Theorem 2.2.26 (Weingarten Gram is NatCast)**.** *The complex Weingarten Gram matrix entries are cast from the natural-number valued Weingarten Gram matrix.*

Theorem 2.2.27 (Natural Weingarten Gram Diagonal)**.** *The diagonal entries of the natural Weingarten Gram matrix equal* d^k*.*

Theorem 2.2.28 (Invertibility of Weingarten Gram Matrix)**.** *For* $k \leq d$*, the Weingarten Gram matrix is invertible.*

Definition 2.2.29 (Weingarten Solution Set)**.** The set of solution vectors c to the Weingarten system.

Theorem 2.2.30 (Weingarten Solution Set is Singleton)**.** *For $k \le d$, the solution set of the Weingarten system is a singleton containing $G^{-1}u$.*

Theorem 2.2.31 (Unique Weingarten Solution)**.** *For $k \le d$, the Weingarten system has a unique solution.*

Theorem 2.2.32 (Inverse Solution Form)**.** *For $k \le d$, the unique coefficient vector is given by $c = G^{-1}u$.*

2.3 The Haar Moment Operator

We define the normalized Haar measure on the unitary group and use it to define the moment operator.

Theorem 2.3.1 (Unitary Entry Norm Bound)**.** *Each entry of a unitary matrix $M \in U(d)$ has norm at most* 1*.*

Theorem 2.3.2 (Compactness of the Unitary Group)**.** *The unitary group $U(d)$ is a compact subset of the space of $d \times d$ complex matrices.*

Theorem 2.3.3 (Compact Space Instance of Unitary Group)**.** *The unitary group $U(d)$ is a compact topological space.*

Theorem 2.3.4 (Second Countability of Matrix Space)**.** *The space of $d \times d$ complex matrices has second countable topology.*

Theorem 2.3.5 (Measurable Space Instance of Unitary Group)**.** *The unitary group $U(d)$ carries the Borel σ-algebra.*

Theorem 2.3.6 (Borel Space Instance of Unitary Group)**.** *The unitary group $U(d)$ is a Borel space.*

Theorem 2.3.7 (Haar Measure of Universe is Positive)**.** *The left Haar measure of the entire unitary group is positive.*

Theorem 2.3.8 (Haar Measure of Universe is Finite)**.** *The left Haar measure of the compact unitary group is finite.*

Definition 2.3.9 (Haar Probability Measure)**.** The Haar probability measure μ_H is the left Haar measure on the compact unitary group $U(d)$ normalized to total mass 1.

Theorem 2.3.10 (Haar Probability is Probability Measure)**.** *The Haar probability measure μ_H is indeed a probability measure (has total mass 1).*

Theorem 2.3.11 (Haar Probability is Left Invariant)**.** *The Haar probability measure μ_H is left-invariant under multiplication.*

Definition 2.3.12 (Endomorphism of Matrix)**.** The representation of a matrix $U \in \mathrm{Mat}_{d\times d}(\mathbb{C})$ as a linear endomorphism of $\mathbb{C}^d$.

Theorem 2.3.13 (Multiplicativity of endOf)**.** *The map 'endOf' satisfies endOf(AB) = endOf$(A) \circ$ endOf(B).*

Theorem 2.3.14 (Identity Matrix endOf)**.** *The map 'endOf' maps the identity matrix to the identity endomorphism.*

Theorem 2.3.15 (Diagonal Action Composition)**.** *The diagonal action is multiplicative: $g^{\otimes k} \circ h^{\otimes k} = (g \circ h)^{\otimes k}$.*

Theorem 2.3.16 (Diagonal Action Identity)**.** *The diagonal action of the identity endomorphism is the identity on the tensor space.*

Definition 2.3.17 (Conjugation Action)**.** For $O \in \mathrm{End}(V^{\otimes k})$ and $U \in U(d)$, the conjugation action is:

$$U^{\otimes k} O (U^{\dagger})^{\otimes k} \tag{2.5}$$

Theorem 2.3.18 (Unitary Diagonal Action Left Inverse)**.** *For any unitary matrix U, $(U^{\dagger})^{\otimes k} \circ U^{\otimes k} = id$.*

Theorem 2.3.19 (Unitary Diagonal Action Right Inverse)**.** *For any unitary matrix U, $U^{\otimes k} \circ (U^{\dagger})^{\otimes k} = id$.*

Theorem 2.3.20 (Continuity of Conjugation Action Entries)**.** *For any operator $O \in End(V^{\otimes k})$, the matrix entries of the conjugation action $actOn(O, U)$ are continuous functions of $U \in U(d)$.*

Theorem 2.3.21 (Integrability of Conjugation Action Entries)**.** *Each matrix entry of the conjugation action is integrable with respect to the Haar probability measure.*

Definition 2.3.22 (Haar Moment Matrix)**.** The matrix formed by integrating the entries of the conjugation action over the unitary group.

Definition 2.3.23 (Haar Moment Operator)**.** The Haar moment operator $\mathbb{E}_{U\sim\mu_H}[U^{\otimes k} O (U^{\dagger})^{\otimes k}]$, defined as the operator whose matrix in the computational basis is the Haar average of the conjugation action.

Theorem 2.3.24 (Matrix of Haar Moment Operator)**.** *The matrix representation of the Haar moment operator matches the Haar moment matrix.*

Theorem 2.3.25 (Pointwise Trace Invariance under Conjugation)**.** *The trace against $V_d^{\dagger}(\sigma)$ is invariant under unitary conjugation of the operator.*

Theorem 2.3.26 (Trace Invariance of Moment Operator)**.** *The moment operator preserves the traces:*

$$Tr(V_d^{\dagger}(\sigma)\mathbb{E}[U^{\otimes k} O (U^{\dagger})^{\otimes k}]) = Tr(V_d^{\dagger}(\sigma) O) \tag{2.6}$$

Theorem 2.3.27 (Translation of Conjugation Action)**.** *Conjugating the action by a unitary matrix W corresponds to left translating the variable in the conjugation action: $W^{\otimes k}(actOn(O, U))(W^{\dagger})^{\otimes k} = actOn(O, WU)$.*

Theorem 2.3.28 (Unitary Invariance of Haar Moment Operator)**.** *The Haar moment operator is invariant under conjugation by any unitary matrix.*

Theorem 2.3.29 (Unitary Commutativity of Moment Operator)**.** *The moment operator commutes with $W^{\otimes k}$ for any unitary matrix $W \in U(d)$.*

We extend this to all matrices using density.

Theorem 2.3.30 (Orthonormal Basis from Matrix Columns)**.** *An orthonormal family can be constructed from the columns of a matrix N when normalized.*

Theorem 2.3.31 (Spectral Decomposition of Hermetian Matrix). *Any positive semidefinite Hermetian matrix $M^\dagger M$ can be diagonalized by a unitary matrix.*

Theorem 2.3.32 (Conjugate Transpose Multiplicative Spectral Relation). *If $M^\dagger M = QDQ^\dagger$, then $(MQ)^\dagger(MQ) = D$.*

Theorem 2.3.33 (Unitary Matrix from Orthonormal Basis). *An orthonormal basis can be assembled into a unitary matrix.*

Theorem 2.3.34 (Column Normalization Relation). *The unitary matrix times the diagonal square root matrix recovers the normalized columns.*

Theorem 2.3.35 (SVD Existence). *Any complex matrix can be written as $M = U_1DU_2$ with $U_1, U_2 \in U(d)$ and diagonal D.*

Theorem 2.3.36 (Matrix of Diagonal Diagonal Action). *The matrix representation of the diagonal action of a diagonal matrix is diagonal in the tensor basis.*

Theorem 2.3.37 (Diagonal Matrix Unitary Criterion). *A diagonal matrix with entries on the unit circle is unitary.*

Theorem 2.3.38 (Identity extension from Torus). *An identity of index-monomials holding on the unit torus holds on all inputs.*

Theorem 2.3.39 (Matrix of Composition). *The matrix representation map is multiplicative: $toEndMatrix(f \circ g) = toEndMatrix(f)toEndMatrix(g)$.*

Theorem 2.3.40 (Entrywise Commutation with Diagonal Action). *An operator commuting with all unitary diagonal actions commutes entrywise with all diagonal actions.*

Theorem 2.3.41 (Commutation with all Diagonal Actions). *An operator commuting with all unitary diagonal actions commutes with the diagonal action of any diagonal matrix.*

Theorem 2.3.42 (Unitary Commutant implies Centralizer). *An operator that commutes with all unitary diagonal actions commutes with all diagonal actions.*

Theorem 2.3.43 (Moment Operator lies in Permutation Span). *The Haar moment operator lies in the span of the permutation operators:*

$$\mathbb{E}_U[U^{\otimes k}O(U^\dagger)^{\otimes k}] \in Span(\mathcal{P}_{d,k}) \tag{2.7}$$

Theorem 2.3.44 (Weingarten Moments for Haar Integral). *The Haar moment operator is a linear combination of the permutation operators whose coefficients solve the Weingarten system.*